Answer the following:

10 solution

$$\text{Rate} = k[A]^2[B][L]^{3/2} \quad \text{-----(1)}$$

(*i*) when $[L] = [4L]$

$$\text{Rate} = k[A]^2[B][4L]^{3/2}$$

$$\text{Rate} = 8\left(k[A]^2[B][L]^{3/2}\right) \text{-----(2)}$$

Comparing (1) and (2) ; rate is increased by 8 times.

(*ii*) when $[A] = [2A] \; and \; [B] = [2B]$

$$\text{Rate} = k[2A]^2[2B][L]^{3/2}$$

$$\text{Rate} = 8\left(k[A]^2[B][L]^{3/2}\right) \text{-----(3)}$$

Comparing (1) and (3) ; rate is increased by 8 times.

(*iii*) when $[A] = \left[\frac{A}{2}\right]$

$$\text{Rate} = k\left[\frac{A}{2}\right]^2[B][L]^{3/2}$$

$$\text{Rate} = \left(\frac{1}{4}\right)\left(k[A]^2[B][L]^{3/2}\right) \text{-----(4)}$$

Comparing (1) and (4) ; rate is reduced to ¼ times.

(*iv*) when $[A] = \left[\frac{A}{3}\right] \; and \; [L] = [4L]$

$$\text{Rate} = k\left[\frac{A}{3}\right]^2[B][4L]^{3/2}$$

$$\text{Rate} = \left(\frac{8}{9}\right)\left(k[A]^2[B][L]^{3/2}\right) \text{-----(5)}$$

Comparing (1) and (5) ; rate is reduced to 8/9 times.

11. solution

Let us consider the dimerisation of a monomer M

$$2M \longrightarrow (M)_2$$

$$\text{Rate} = k[M]^n$$

Given that n=2 and [M] = 0.05 mol L^{-1}

Rate = 7.5 X 10^{-3} mol $L^{-1}s^{-1}$

Answers.indd 251 22-12-2022 19:40:28